

Assessment Test3
Embedded Systems -B9
Advance C

Date: 27-08-2026

Marks : 50M

Q1. Problem Statement

20M

Write a C program to implement a simple Sensor Status Processing System for an embedded device.

The system monitors 2 sensors and stores the following information:

- **Sensor name**
- **Sensor ID**
- **Sensor temperature**
- **Sensor status**

Store the sensor names, IDs, temperatures, and status values using appropriate arrays.

- **Display the details of all sensors.**
- **Implement a function that updates the status byte by:**
 - **Clearing the Error bit.**
 - **Setting the Data Ready bit.**
 - **Setting the Enabled bit.**
 - **Keeping the sensor mode unchanged.**
- **Display the updated status of each sensor after performing the required bit manipulations.**
- **Bit 0 → Sensor Enabled**
- **Bit 1 → Sensor Error**
- **Bit 2 → Data Ready**
- **Bit 3 → Calibration Required**
- **Bits 4-7 → Sensor Mode**
- **Sensor 1:**

Name : Temperature

ID : 101

Temperature : 32.5

Status : 0x25

- Sensor 2:

Name : Pressure

ID : 102

Temperature : 45.2

Status : 0x13

Q2 Dynamic Number Analyzer

10M

Problem Statement

Write a C program that accepts an unknown number of integers from the user.

Example:

Enter number of elements: 6

Enter elements:

12 7 25 7 40 12

The program should:

- 1. Dynamically allocate memory for the elements.**
- 2. Find the largest and smallest values.**
- 3. Find all duplicate values.**
- 4. Release all dynamically allocated memory.**

The program must use:

- **Dynamic memory allocation**
- **Arrays and pointers**
- **User-defined functions**
- **Pointer arithmetic**
- **Proper memory deallocation**

Q3— Command-Line String

10M

Check whether the string is a palindrome. If it is, display "Palindrome"; otherwise, display "Not Palindrome" and print the reverse of the string.

Note: Use pointer notation while traversing the string wherever appropriate.

Example execution

./a.out level apple madam computer

Expected output

String: level

Palindrome: Yes

Short answer

Q4. Double Pointer **5M**

- a) Write the definition of a double pointer in C.**
- b) Write the syntax for declaring a double pointer.**
- c) Write a C program to demonstrate the use of a double pointer.**

Q5 Dynamic Memory Allocation **5M**

- a) Define dynamic memory allocation in C.**
- b) Write the names and syntax of the dynamic memory allocation functions**
